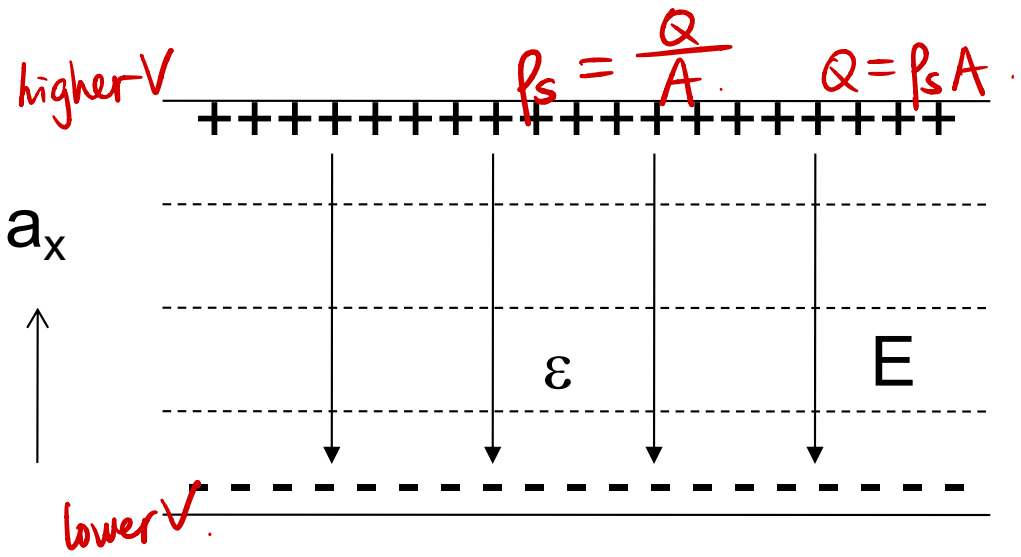
ECE 329
Lectures 10

Capacitance and Conductance

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Parallel-plate Capacitor



x=d, Q

Equipotential lines

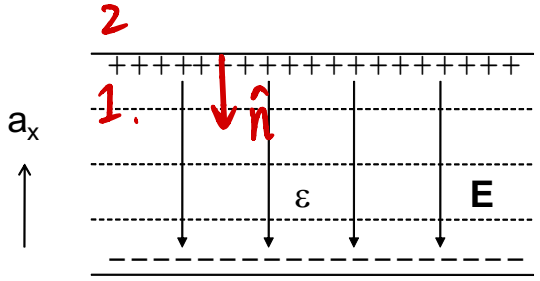
x=0, -Q

Capacitance is: $C = Q/V$

V → E → D → ρ_s → $Q = \rho_s A$

Steps to Find Capacitance

Given V, unknown Q



ε is constant and
ρ=0 in between

- Laplace Equation $\nabla^2 V = 0$
- Find V using boundary conditions

$$V(z) = V_0 \frac{x}{d} \quad [V]$$

- Find **E** using $\vec{E} = -\nabla V = -\frac{\partial}{\partial x} V \hat{x}$
- Find **D** using $\vec{D} = \varepsilon \vec{E}$

$$\vec{E} = -\frac{V_0}{d}\hat{x} \quad [V/m]$$

$$\vec{D} = -\frac{\varepsilon_0 V_0}{d}\hat{x} \quad [C/m^2]$$

- Get surface charge density on one conductor using BC

$$\rho_s = \vec{a}_n \bullet (D_{n1} - D_{n2})$$

Top $\rho_s = (-\hat{x}) \cdot \left(-\frac{\varepsilon_0 V_0}{d}\hat{x} - 0\right)$

$$\rho_s = +\frac{\varepsilon_0 V_0}{d} \quad [C/m^2]$$

- Total charge $Q = (Area)(\rho_s)$

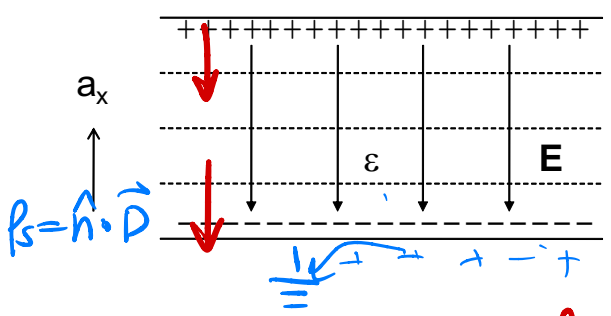
$$Q = \frac{\varepsilon_0 A V_0}{d} \quad [C]$$

- Capacitance $C = Q/V$

$$C = \frac{\varepsilon A}{d} \quad [F]$$

Steps to Find Capacitance

Given Q, unknown V



ε is constant and ρ=0 in between

- Find $\vec{D}$ using $\rho_s \vec{D} = \frac{\rho_s}{2} \mathrm{sgn}(x)\, \hat{x}$ $\quad \rho_s = \frac{Q}{A}$
- Find $\vec{E}$ using $\vec{D} = \epsilon \vec{E}$
- Calculate voltage drop

$$V = \int_0^d \vec{E} \cdot d\vec{l} \quad = E \cdot d$$

- Capacitance

$$C = Q/V$$

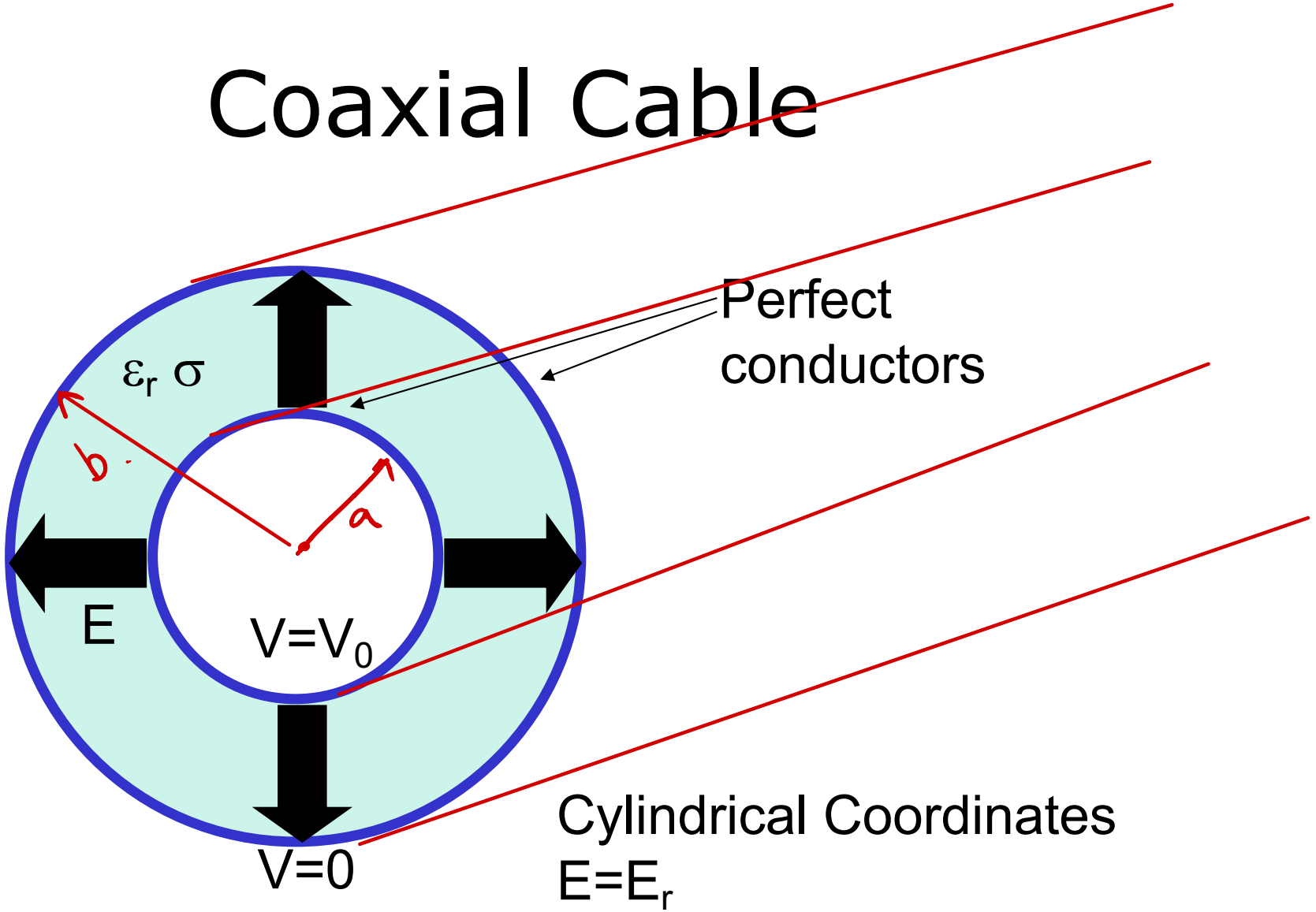
$\vec{D} = \rho_s(-\hat{x}) = -\frac{Q}{A}\hat{x}$ [C/m²]

$\vec{E} = -\frac{Q}{A\varepsilon}\hat{x}$ [V/m]

$V = \frac{Qd}{A\varepsilon}$ [V]

$C = \frac{\varepsilon A}{d}$ [F]

Coaxial Cable



Cylindrical Coordinates

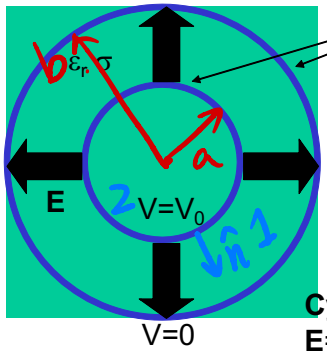
$E=E_r$

z direction along length of the cable

Coaxial Cable

V → E → D → ρ_s → Q

Given V, unknown Q



Cylindrical Coords
E=E_r
z direction along length of the cable

per unit length $C = \frac{C}{L}$

$= \frac{2\pi\varepsilon}{\ln(b/a)}$ [F/m]

① $\nabla^2 V = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) = 0$

$\left(r\frac{\partial V}{\partial r}\right) = A$ const.

$\frac{\partial V}{\partial r} = \frac{A}{r}$

$V(r) = A\ln(r) + B$ (A, B: const.)

$V(a) = V_0$, $V(b) = 0$

$V(r) = \frac{V_0 \ln(r/b)}{\ln(a/b)}$ [V]

② $\vec{E} = -\nabla V = -\frac{\partial V}{\partial r}\hat{r} = -\frac{V_0}{\ln(a/b)}\cdot\frac{1}{r}\hat{r}$ [V/m]

③ $\vec{D} = \varepsilon\vec{E} = \frac{V_0\varepsilon}{\ln(b/a)}\frac{1}{r}\hat{r}$ [C/m²]

④ $\rho_s = \hat{n}\cdot(\vec{D}_1 - \vec{D}_2) = \hat{r}\left(\frac{V_0\varepsilon}{\ln(b/a)}\cdot\frac{1}{r}\hat{r} - 0\right)\Big|_{r=a}$

Inner conductor

$\rho = \begin{cases} \varepsilon V_0/(a\ln(b/a)), r = a \\ -\varepsilon V_0/(b\ln(b/a)), r = b \end{cases}$

$= \frac{V_0\varepsilon}{\ln(b/a)}\cdot\frac{1}{a}$ at r=a [C/m²]

⑤ $Q = \rho_s \cdot A = \frac{V_0\varepsilon\cdot 2\pi L}{\ln(b/a)}$ [C]

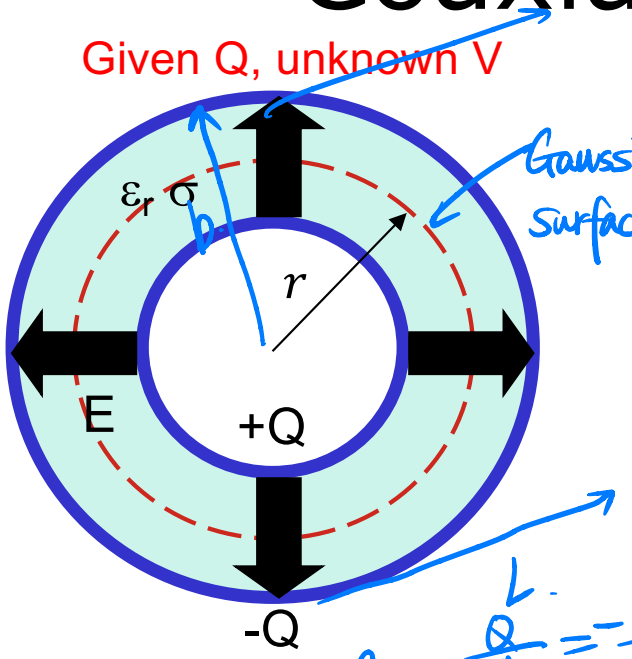
length L. $A = 2\pi a L$

⑥ $\frac{Q}{V} = C = \frac{2\pi\varepsilon L}{\ln(b/a)}$ [F]

length of L

Coaxial Cable Capacitance

(Review at home)



Gauss's law on cylindrical surface with a radius $r > a$

$$\epsilon \oint_S \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{S}} = Q \qquad \epsilon E_r 2\pi r L = Q$$

$$E_r = \frac{Q}{2\pi\epsilon r L}$$

Voltage drop from inner to outer conductor is

$$V = -\int_b^a \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}} = \int_a^b E_r dr = \int_a^b \frac{Q}{2\pi\epsilon r L} dr$$

$$= \frac{Q}{2\pi L\epsilon}\int_a^b \frac{dr}{r} = \frac{Q}{2\pi L\epsilon}\ln\left(\frac{b}{a}\right)$$

Cylindrical Coordinates
E=E_r
z direction along length of the cable

Use $C = Q/V$

Capacitance of the coax segment of length L:

$$C = \frac{2\pi\epsilon L}{\ln(b/a)} \quad [\text{F}]$$

Per unit length capacitance

$$\mathcal{C} = \frac{C}{L} = \frac{2\pi\epsilon}{\ln(b/a)} \quad [\text{F/m}]$$

Challenge Question: Coaxial cable capacitance

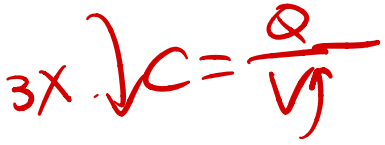
- Consider a coax with ($\epsilon_{old} = 3\epsilon_0$) for $a < r < b$. If we remove the dielectric and replace it with free space ($\epsilon_{new} = \epsilon_0$) but **keep the same amount of charge** on each PC, which is **false**?

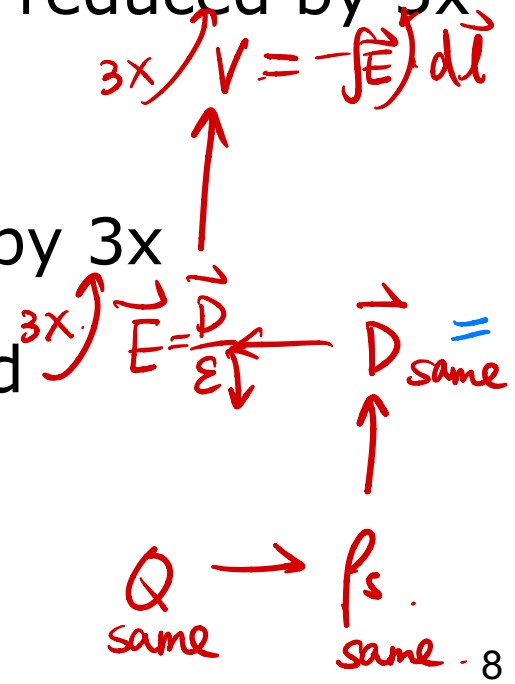
(a) the voltage across the capacitor is reduced by 3x

(b) the capacitance is reduced by 3x

(c) the E-field for a<r<b is increased by 3x

(d) the D-field for a<r<b is unchanged





Challenge Question: Coaxial cable capacitance

- Consider a coax with ($\epsilon_{old} = 3\epsilon_0$) for $a < r < b$. If we remove the dielectric and replace it with free space ($\epsilon_{new} = \epsilon_0$) but **keep the same voltage** across the coax, which is **false**:

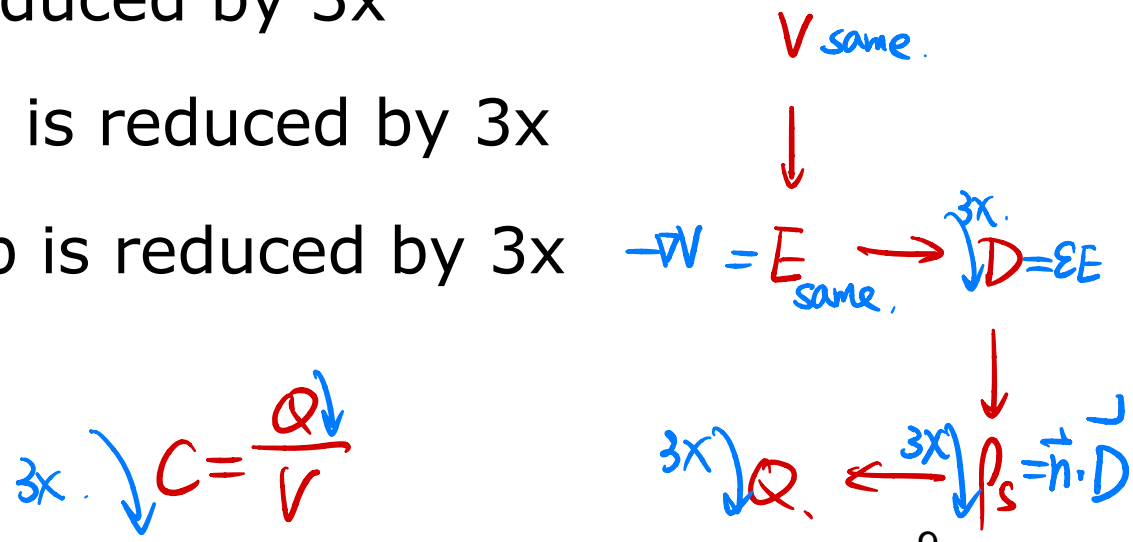
$\varepsilon \downarrow 3x$.

(a) the charge across the capacitor is reduced by 3x

(b) the capacitance is reduced by 3x

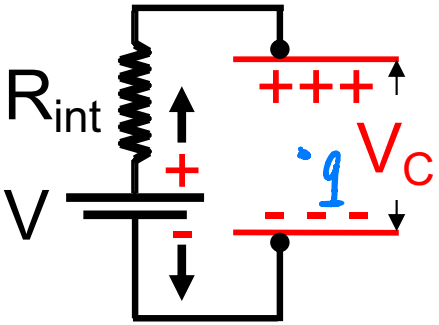
(c) the E-field for a<r<b is reduced by 3x

(d) the D-field for a<r<b is reduced by 3x



Energy Stored in a Capacitor

- The work by the battery to move a charge dq from the bottom to the top plate is:



$$dW = V_c dq = \frac{q}{C} dq$$

$C = \frac{Q}{V}$

V_C is instantaneous voltage on the capacitor.
It starts at zero and increases to V

- Thus the total stored energy while charging is:

$$W = \int_0^Q dq \frac{q}{C} = \frac{Q^2}{2C} = \frac{1}{2} CV^2$$

$$W_{\substack{\text{fully} \\ \text{charged}}} = \frac{1}{2} CV^2$$

because $V_c = V$
after charging

Stored energy of a capacitor

- Stored energy is: $C = \epsilon_o \frac{A}{d}$ $V = E_x d$

$$W = \frac{1}{2}CV^2 = \frac{1}{2}\epsilon_o |E_x|^2 Ad$$ Fully charged capacitor

- Stored energy can be written is

$$W = \frac{1}{2}\epsilon_o |E_x|^2 \times Ad$$

$$\mathrm{w} = \frac{1}{2}\epsilon |E_x|^2 = \frac{1}{2}\epsilon \vec{E} \cdot \vec{E}$$

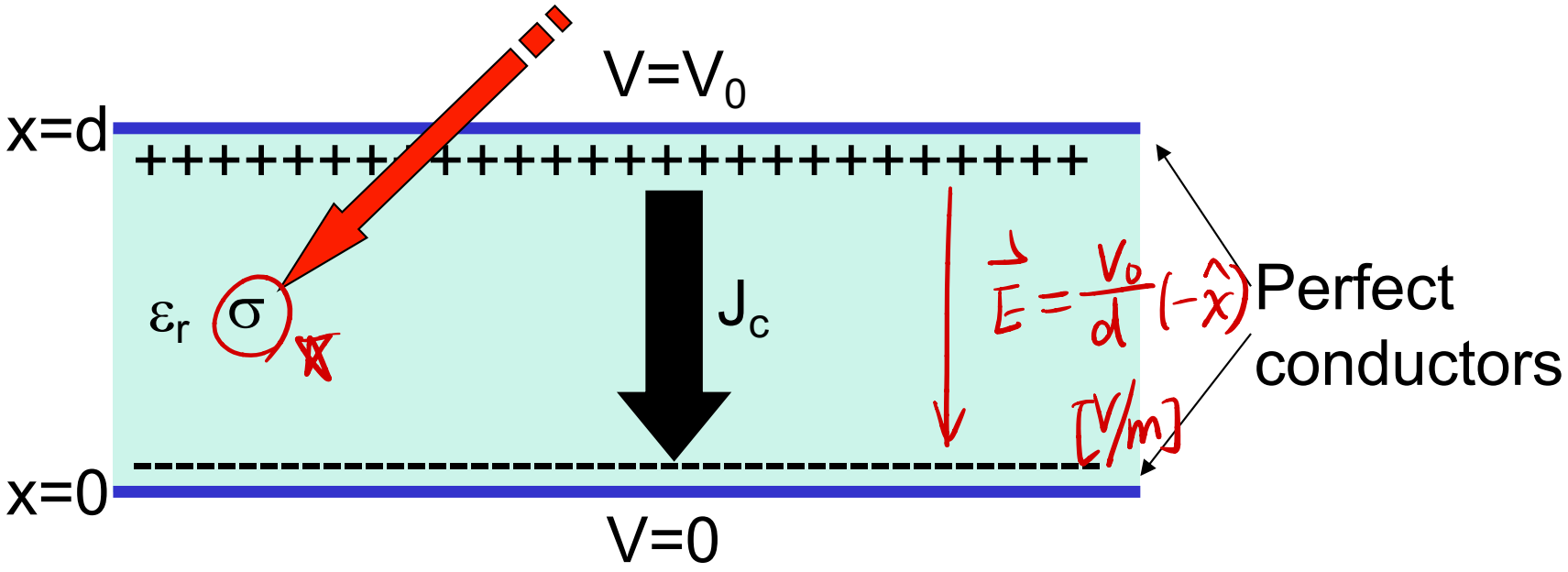
stored electrostatic energy per unit volume $[W/m^3]$

$$Vol = Ad$$

volume of the capacitor

Conductance $G = \frac{1}{R} = \frac{I}{V}$

Medium in capacitor now has conductivity



Now a current can flow x=d to x=0

$$\vec{J} = \sigma\vec{E} = \sigma\left(\frac{V_0}{d}\right)(-\hat{x})$$

Current density (A/m^2)

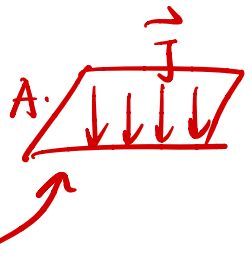
Conductance

conductor current.

$$G = \frac{|I_c|}{V_0} \quad \text{or } I = GV$$

Conductance
Units: Siemens

For the parallel plate capacitor



$$\vec{J} = \sigma\vec{E} = \sigma\left(\frac{V_0}{d}\right)(-\hat{x}) \qquad I = JA$$

$$\sigma \cdot \frac{V_0}{d} \cdot A \,/\, V_0$$

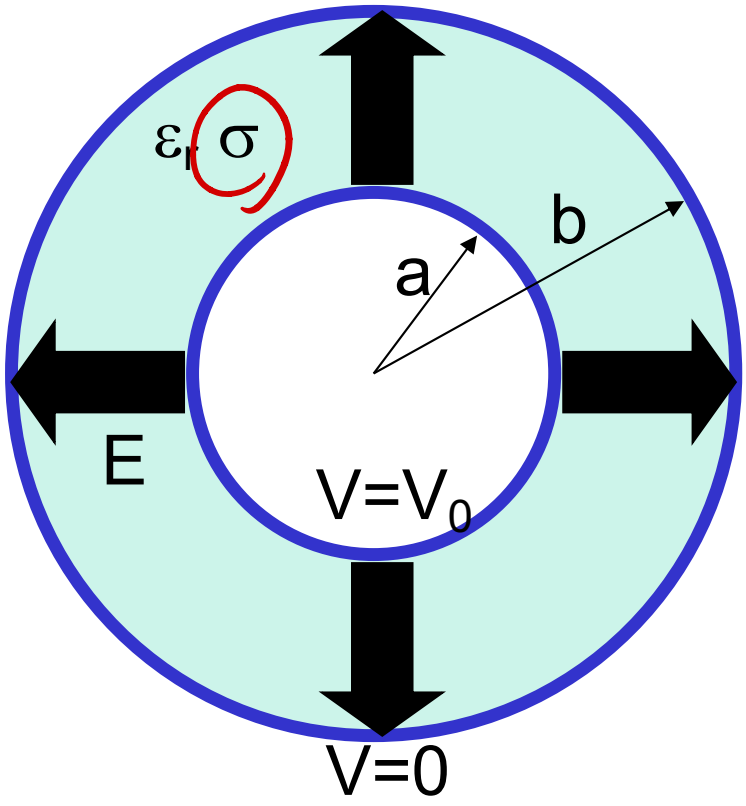
Conductance $\boxed{G = \frac{\sigma A}{d}}$ Capacitance $\boxed{C = \frac{\varepsilon A}{d}}$

$G = \sigma \cdot$ Geometric Factor $\qquad C = \varepsilon \cdot$ G.F.

Relation between conductance and capacitance

$$\boxed{G = \frac{\sigma}{\epsilon} C} \qquad \frac{G}{C} = \frac{\sigma}{\varepsilon}$$

Conductance/Length for Coaxial Cable

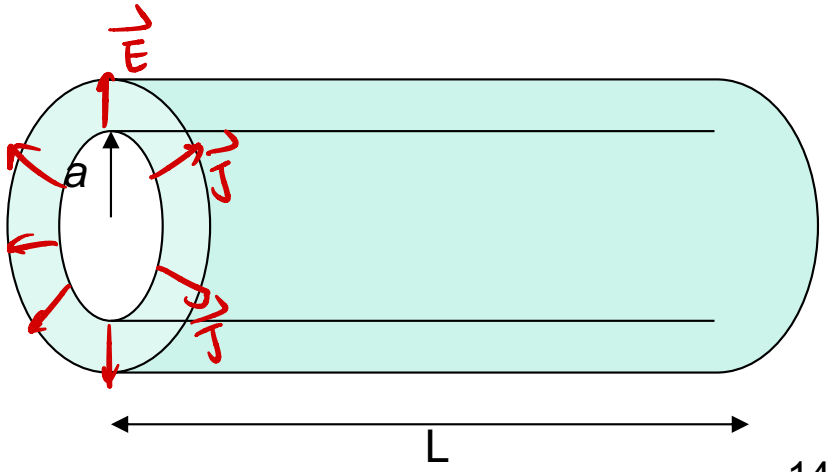


$$\vec{E} = \frac{V_0}{\ln(b/a)}\left(\frac{1}{r}\right)\vec{a}_r$$

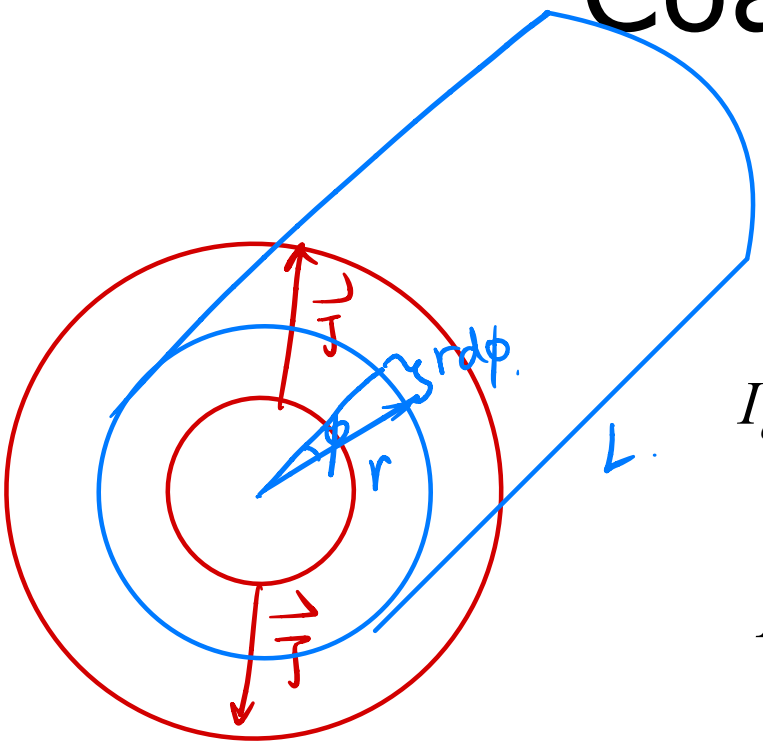
so

$$\vec{J}_c = \sigma\vec{E} = \frac{\sigma V_0}{\ln(b/a)}\left(\frac{1}{r}\right)\vec{a}_r \qquad \left[\frac{A}{m^2}\right]$$

What is radial current (I_c) for a fixed length of the cable?



Conductance/Length of Coaxial Cable



$$I_c = \int \vec{J} \bullet d\vec{S} \qquad [A]$$

$$I_c = \int_{\phi=0}^{2\pi} \int_{z=0}^{L} J_c (r d\phi dz)$$

$$I_c = \frac{2\pi\sigma V_0 L}{\ln(b/a)}$$

$$G = \frac{I_c}{V_0} = \frac{2\pi\sigma L}{\ln(b/a)} \qquad [Siemens]$$

Conductance per unit length

$$G = \frac{G}{L} = \frac{2\pi\sigma}{\ln(b/a)} \qquad [S/m]$$

$$C = \frac{2\pi\varepsilon}{\ln(b/a)} \qquad [F/m]$$

Steps for Finding Conductance

- Find Electric Field
- Find Conduction Current Density (A/m^2)

$$\vec{J}_c = \sigma \vec{E}$$

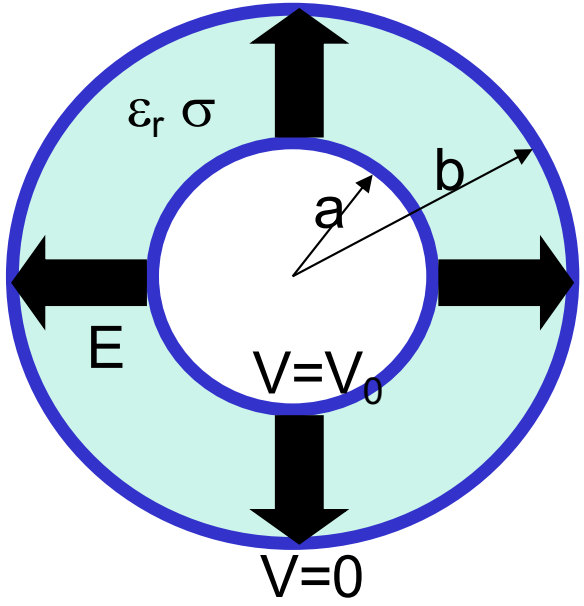
- Conduction Current (A) $I_c = \int \vec{J} \bullet d\vec{S}$
- Conductance $G = \dfrac{I_c}{V_0}$
- Conductance per unit Length

Conductance/Length for Coaxial Cable

(Review at home)

Given V, unknown Q or I

$$V \rightarrow \vec{E} \rightarrow \vec{J}$$



Step 1: Laplace's Equation

$$\nabla^2 V = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) = 0 \qquad a < r < b$$

$$r\frac{\partial V}{\partial r} = constant = A \rightarrow \frac{\partial V}{\partial r} = \frac{A}{r}$$

$$\rightarrow V(r) = A\ln(r) + B$$

Use boundary conditions $V(a) = V_0$ and $V(b) = 0$

$$V = A\ln(r) + B \Rightarrow V = V_0\frac{\ln(b/r)}{\ln(b/a)}$$

Step 2: $$\vec{E} = -\nabla V = -\frac{\partial V}{\partial r}\hat{r} = \frac{V_0}{\ln(b/a)}\left(\frac{1}{r}\right)\hat{r} \quad \left[\frac{V}{m}\right]$$

Step 3: $$\vec{J} = \sigma\vec{E} = \frac{\sigma V_0}{\ln(b/a)}\left(\frac{1}{r}\right)\hat{r} \quad \left[\frac{A}{m^2}\right]$$

Conductance/Length of Coaxial Cable

(Review at home)

$$V \rightarrow \vec{E} \rightarrow \vec{J} \rightarrow I \rightarrow G$$

Step 4:

$$I = \int \vec{J} \cdot d\vec{S} \qquad [A]$$

$$I = \int_{\phi=0}^{2\pi} \int_{0}^{l} \frac{\sigma V_0}{ln(b/a)} \left(\frac{1}{r}\right) (r d\phi dz)$$

$$I = \frac{2\pi\sigma V_0 l}{ln(b/a)}$$

Step 5:

conductance for cable with length L

$$G = \frac{I}{V_0} = \frac{2\pi\sigma l}{\ln(b/a)} \qquad [Siemens]$$

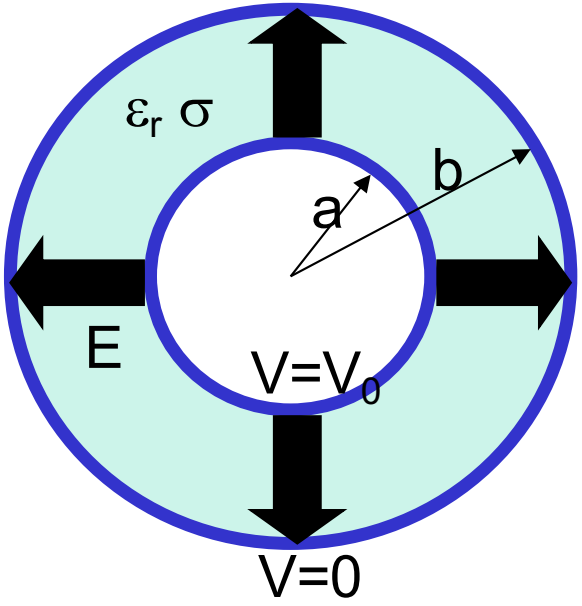
per unit length conductance

$$\mathcal{G} = \frac{G}{l} = \frac{2\pi\sigma}{\ln(b/a)} \qquad [S/m]$$

$$G = \frac{\sigma}{\epsilon} C$$

Relationship between G and C

Conductance/Length for Coaxial Cable

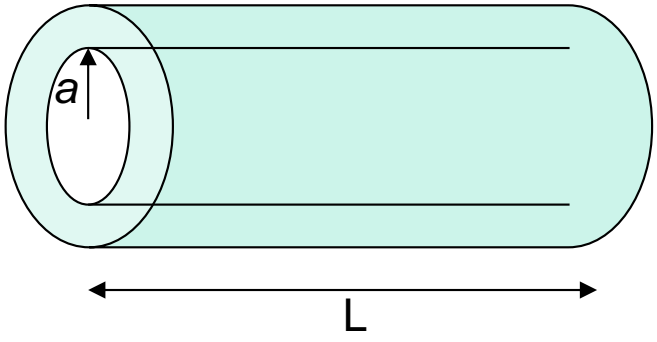


Relationship between G and C

$$G = \frac{\sigma}{\epsilon} C$$

Conductance for cable with length L

$$G = \frac{I}{V} = \frac{2\pi\sigma l}{\ln(b/a)} \quad [Siemens]$$



Per unit length conductance

$$\mathcal{G} = \frac{G}{l} = \frac{2\pi\sigma}{\ln(b/a)} \quad [S/m]$$

Lecture 10 Summary

- Capacitance $C = Q/V_0$
- Conductance $G = |I_c|/V_0$
- Next Up
  - Conductivity and Susceptibility